Module 6A – The Nature of Light



In this Module, we will explore the phenomenon of light.

Optics is

> the study of light in all its forms, both visible and invisible.

A **simple definition** of light is

> electromagnetic radiation.

A **more complex but limited definition** of light is

> a dual transverse, non-mechanical, self-propagating wave that travels at a constant speed in a homogeneous medium.

The **speed of light** in the vacuum of outer space is precisely

> 299,792,458 meters per second,

and this speed is a **law of physics**. It is the fastest any physical entity can go. Nothing material can equal or exceed this speed.

The reason the more complex definition of light is limited is that light has an inherently *contradictory* nature. This contradictory nature of light can be expressed in the form of a question:

> Is light a wave or is it a particle?

A wave by its nature is expansive, whereas a particle (classically speaking) is highly localized. So it would seem that the answer to the above question is one or the other, light is a wave or light is a particle, but not both.

But experiments repeatedly show that light is *both a wave and a particle*. If this is confusing, you are in good and abundant company.

Western Europe's scientific exploration of light began in the 17th century, with the works of Sir Isaac Newton, in England, and Christiaan Huygens in Holland.

Newton, amongst others, thought that light was composed exclusively of particles, which he called corpuscles. Huygens, on the other hand, thought that light was composed exclusively of waves.

Though many scholars throughout history have investigated the nature of light, it is fair to say that from the Renaissance forward, Newton is considered the founder of the

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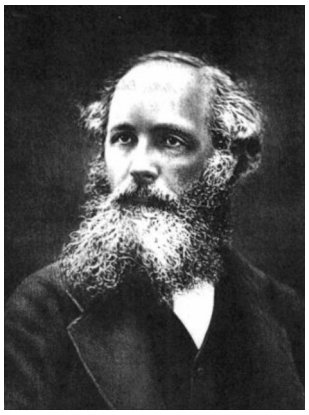
geometric school of light, where light particles' paths are traced as straight lines called *rays*.

Huygens, by comparison, and in the same time frame, is considered the founder of the **physical school** of light, where light waves' paths are drawn as advancing *wave fronts* (joint crests).

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We will use both the ray and wave front models of light in this module, but we will nevertheless assume, for convenience, the primacy of the wave model of light, while acknowledging that certain light phenomena discovered (and ultimately explained) in the late nineteenth and early twentieth centuries can only be explained by the particle model of light.

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Much of the confusion surrounding how to describe light as a wave was (temporarily) put to rest by the inimitable and brilliant Scottish physicist James Clerk Maxwell, whose electromagnetic theory was later condensed by the English physicist Oliver Heaviside into four equations, now known as Maxwell's Equations.

Besides fixing one of the equations, all previously discovered by others, Maxwell predicted the existence of an *electromagnetic wave* whose speed was equal to the already measured speed of light. Maxwell realized that

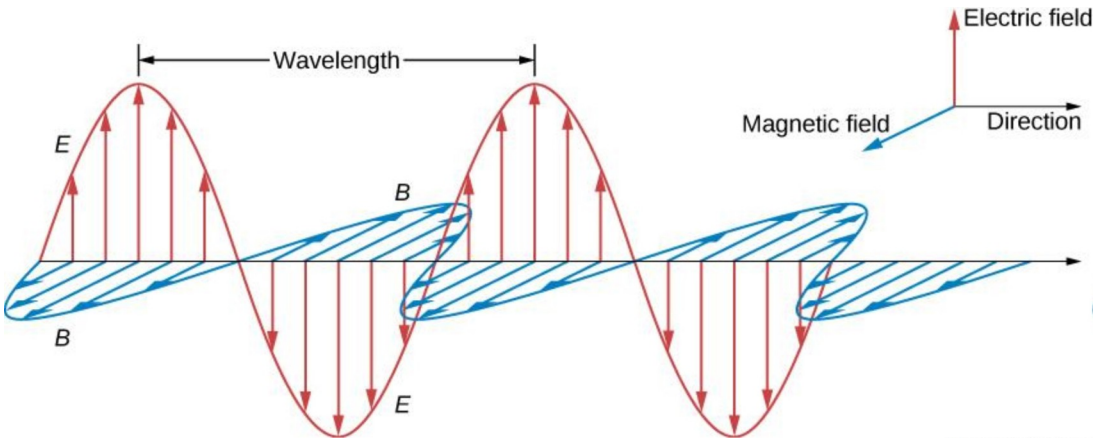
Light itself is an electromagnetic wave.

Maxwell determined (from first principles) that the speed of light (c) was, approximately,

3×10^8 meters/sec = 186,000 miles / sec = 300,000 kilometers/sec

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From a classical perspective, we view light as a non-mechanical, dual transverse, self-propagating, electromagnetic wave:



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Here we see a light wave, generated by a sinusoidally (sine wave like) oscillating electron on an antenna, composed of both a sinusoidal electric field disturbance and a sinusoidal magnetic field disturbance varying at right angles to each other, and to the direction of propagation of the wave (which is to the right).

> We say that the electric field disturbance creates the magnetic field disturbance (in accordance with Ampère–Maxwell Law), and the magnetic field disturbance, in response, creates the electric field disturbance (in accordance with Faraday's Law), back and forth, *ad infinitum*.

This is what is meant by the expression "self-propagating."

Since light's two disturbances are not material in nature, **light** is

> a non-mechanical wave.

Light "does not get tired" in moving from one location to another, even over long distances, since no replenishment of energy is required to sustain the movement of a non-material self-propagating joint disturbance through the apparent emptiness of outer space.

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Just because light "does not get tired" when moving from one location to another does not mean that light does not get *dimmer* the farther away an observer is from a light source.

By "getting dimmer" is meant "losing intensity or brightness (a term of human perception)."

Like sound in air, an ordinary, omnidirectional light source obeys the **Inverse Square Law**:

> The intensity of an ordinary light source decreases with the distance squared.



The **intensity of light** is

> the power of a light source divided by the area upon which the light is incident.

Light intensity is measured in watts per square meter (W/m^2), not lumens, which measure perceived brightness.

Note that the Inverse Square Law can be used *to determine distances*.

This is especially useful in astronomy, where certain stars are known to have an intrinsic brightness at a specific distance. By measuring the star's apparent brightness, astronomers can infer mathematically how far the star must be away from Earth.

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Some light sources *do not* dim much with distance, notably lasers.

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Light is said to be **temporally coherent** if it is

all of one frequency.

Light is said to be **spatially coherent** if

its wave fronts stay in step (crests lined up) across the width of the beam.

Light that is temporally and spatially coherent is simply called **coherent light**.

Laser light is coherent. It is monochromatic, and its beam is highly collimated (its rays are nearly parallel), so its intensity decreases only very slowly with distance.

The Electromagnetic Spectrum

Like sound, a light wave obeys the formula

$$v = f \times \lambda \qquad (10.1)$$

where v is the speed of light in the medium the light is traveling through, f is the frequency of the light being observed, and λ is the wavelength of the light being observed.

The wavelengths of light that ***humans can see*** range from approximately 700 nanometers to 400 nanometers. A nanometer (nm) is 10^{-9} meters, a billionth of a meter.

Note: Some humans can see wavelengths shorter than 400 nm and longer than 700 nm.

Visible light near 700 nm is perceived to be the color dark red, and visible light near 400 nm is perceived to be the color dark violet.

Note: Color is a ***perception*** of a visible light wave, not an inherent property of that light wave.

All other wavelengths (and corresponding frequencies) of light are invisible, meaning they exist, but humans cannot perceive them. Since there's no known frequency variation limitations for light, we can construct an **electromagnetic spectrum**, or range array of light frequencies by type.

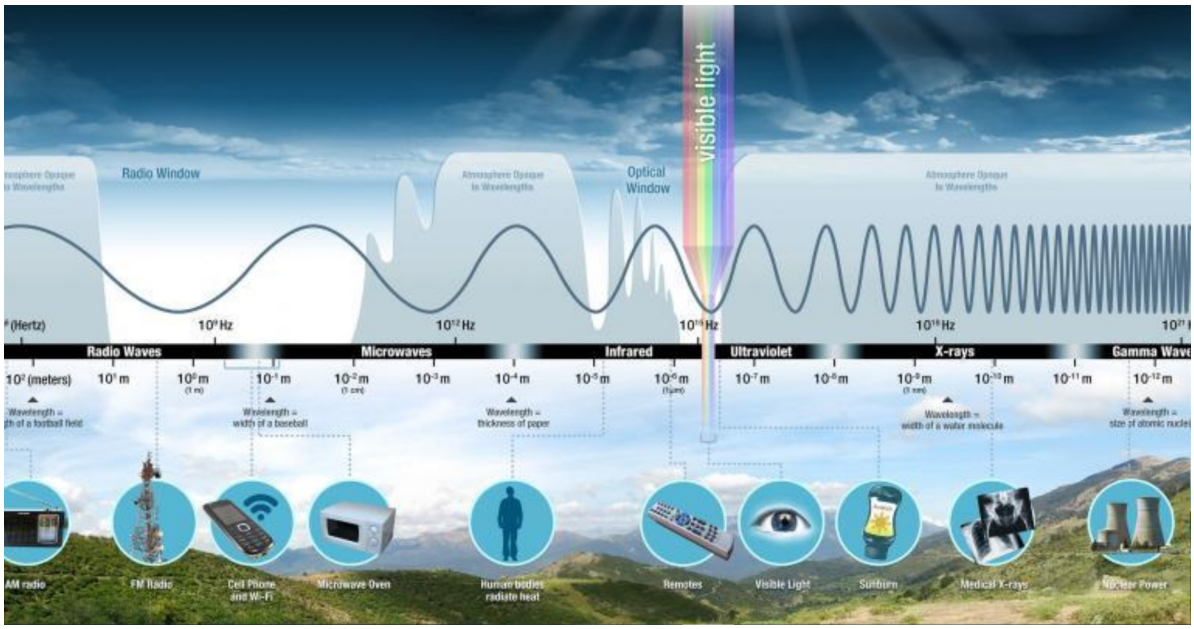
In general, **Ranges** of the electromagnetic spectrum, in order of increasing frequency, can be remembered as

radio, tv, microwave, infrared, visible, ultraviolet, x-ray, and gamma ray.

The highest frequencies are called gamma rays. Despite the name, cosmic rays are not light at all: they are high-energy charged particles, mostly protons, arriving from space.

An image follows, showing the electromagnetic spectrum by types and applications.

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How Light Interacts With Matter

The journey of light begins with matter and ends with matter. On its journey, light interacts with matter in one of **three ways**:

reflection, transmission and absorption.

Reflection of light is

the bouncing of light off matter.

Unless an object is emitting light in the visible range of frequencies, you can only see objects that reflect visible light that subsequently enters your eyes.

You cannot see light passing by you. Light is not a thing that light reflects off of.

Transmission of light is the

movement of light through something (solid, liquid or gas).

Transmission can be ***transparent*** (clear, with no diffusion), or ***translucent*** (diffusion occurs). Diffusion means "spreading about in random directions."

When a material does not allow movement of light through it, we say it is ***opaque***.

Absorption of light in matter involves

the transmutation of the light's energy into thermal energy in the matter, often with the subsequent re-emission of new light out of the matter at a lower frequency.

The re-emitted light is typically at a lower energy, because some of the energy of the initially incident light is dissipated in the movement of atomic particles in the matter.

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Classical Phenomena of Light

Light *cannot* be described in the same way as an object that can be observed -- with light -- at rest.

Light is always moving with respect to the observer: this is a consequence of the constancy of the speed of light. We cannot "catch up" to light and look at it (with light).



We can however discover hints as to light's intrinsic nature, by looking at well-known phenomena of light.

These phenomena are divided into two groups:

classical phenomena (discovered pre-twentieth century), and

modern phenomena (discovered in the twentieth century, or later).

Some phenomena "prove" light is a particle, whereas other phenomena "prove" light must be a wave, hence light's contradictory nature.

For now, we will look only at the **classical phenomena** of light.

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Classical light phenomena include the following:

Reflection, which is

the bouncing of light off matter.

We will learn a law about light's reflection, and do a lab activity on reflection later.

Refraction, which is

the bending of light as it passes from one optical medium to another of a different optical density.

By optical medium, we mean a substance that is transparent or translucent.

We will learn a law about refraction, and do a lab activity on refraction later.

Dispersion, which is

the breaking up of white light into its component colors.

Note: Dispersion is an inherently refractive phenomenon.

We see dispersion when using prisms or observing rainbows. Newton discovered that white light is not a separate color, but a combination of colors.

Diffraction, which is

the bending of light around corners.

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We will learn a law about diffraction, and do a lab activity on diffraction later.

And the **Doppler Effect**, which is

> the observance of a shifting of a light wave's frequency, because of relative motion between the source of the light and the observer of the light.

Note: The particle model of light can only adequately explain reflection, but the wave model of light can explain all of these classical light phenomena.

Note: All these classical light phenomena occur also for sound in air, even though light is a non-mechanical wave, and sound in air is a mechanical wave.

Geometric Optics

Geometric Optics is the school of optics that models light as a

> stream of particles moving in straight lines when not bouncing off matter, or going through matter.

Geometric Optics is particularly suitable to use when discussing image formation using mirrors and lenses.

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We first look at image formation using mirrors.

Mirrors are exclusively reflective. They are typically very smooth of surface and highly reflective (having a high albedo).

This means that mirrors exhibit **specular reflection**, that is

> image integrity is maintained in reflection.

Image integrity means that you can

> see an object in reflection and still be able to identify it.

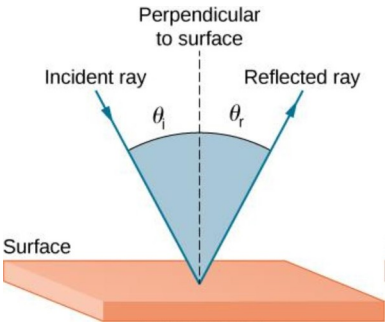
Image integrity does not mean that a reflection cannot exhibit distortion: flat mirrors reflect without distortion, but curved mirrors reflect with distortion.

In contrast to specular reflection, **diffuse reflection** occurs

> when light reflects in random directions off a less than shiny and less than smooth surface.

Nevertheless, light rays obey the same law of reflection, whether the light is reflecting specularly or diffusely.

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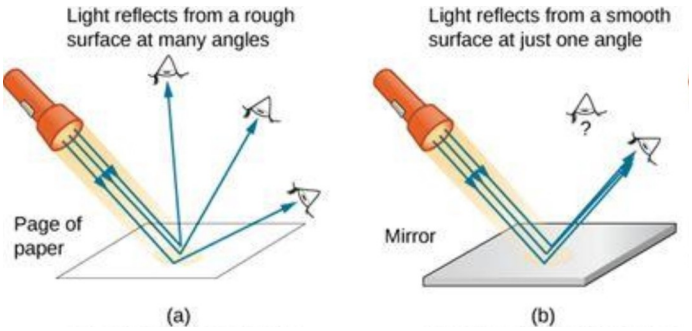
The Law of Reflection states that

the angle made by the incoming (or incident) ray, and the normal (or perpendicular) line up from the point of impingement, equals the angle made by the outgoing (or reflected) ray and the normal (or perpendicular) line up from the point of impingement.

Put in simpler terms,

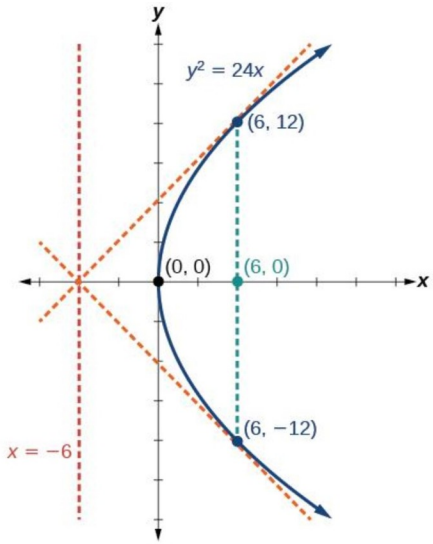
the angle of incidence equals the angle of reflection ($\theta_i = \theta_r$).

Note: The incident ray, the reflected ray, and the normal line must all fall within the same plane.



Returning now to specular and diffuse reflection, we see that image integrity is maintained on a mirrored surface, because parallel rays from an image will be reflected at the same angle (b), but will be reflected at different angles on a microscopically rough surface (a).

Consider now a mirror whose cross-section is shaped like a parabola:



The mirror itself, in three dimensions, is generated by "spinning" this parabolic cross-section around the axis of symmetry (here the x axis). The mirror shape is called a paraboloid, but here we will call it a **parabolic mirror**.

The peculiar thing about this mirror is that all rays coming in from the right that are parallel to the axis of symmetry (called **paraxial rays**) will, upon obeying the law of reflection, end up at the light green point (6,0) called the **focal point**.

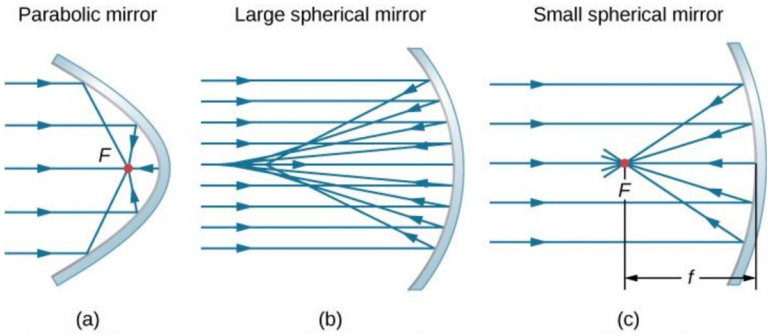
Conversely, all rays coming from the focal point and reflecting off the parabolic mirror will exit paraxially (and parallel to each other) to the right of the mirror.

When, for example, rays from the Sun (which are all parallel) strike a parabolic mirror paraxially, they all come together at the mirror's focal point. As a result, the mirror's focal point becomes extremely hot (think solar cooker.

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Consider now an approximation of a parabolic mirror, called the small concave spherical mirror:



In image (a) we see paraxial rays coming together at a parabolic mirror's focal point F. In image (c) we see paraxial rays coming together almost perfectly at the small (concave) spherical mirror's focal point, again called point F.

In image (b), we see the limitation of using a spherical mirror as an approximation for a parabolic mirror. The mirror is too large, and the paraxial rays do not come together well at a single focal point.

Note: The mirror in image (c) is called concave, because the mirror is silvered (shiny) on the left side of the mirror, and all the rays of light shown are coming in from the left.

Note: Point F is on the **axis of symmetry** of the concave spherical mirror. Twice the distance out on that same axis is point R, which is the center of the sphere of which the mirror is a part.

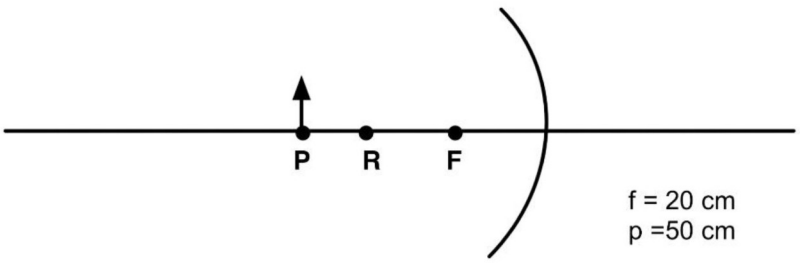
We say that

$$r = 2f, \tag{10.2}$$

where r is the radius of curvature of the spherical mirror, and f (shown above) is the focal length of the spherical mirror.

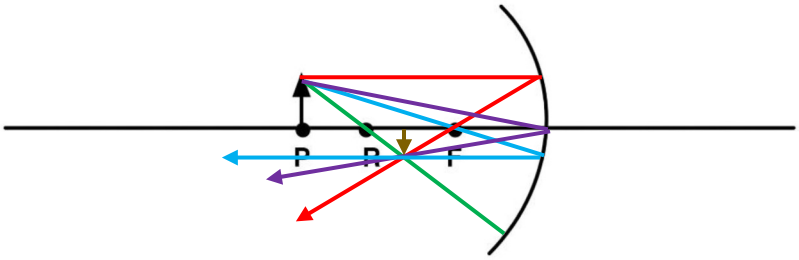
We are now ready to do some image formation, using ray tracing.

Consider the concave mirror system shown below. The object is placed at point P, a distance p away from the mirror's vertex, or center point (the object distance), where R is the center of the spherical mirror segment, and F is the focal point of spherical mirror.



Since f is given, we know that r is twice as much, or 40 cm. The distance p is the **object distance**. We can construct an image using up to four rays:

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Ray 1 is the red ray, and comes in paraxially, so reflects off the mirror through the focal point F. Ray 2 is the light blue ray, and comes in through the focal point, so reflects paraxially. Ray 3 is the green ray, a radius of curvature ray going through point R, so it reflects back on itself. Ray 4 is the purple ray, a symmetric ray, so it is easy to see and draw its angle of reflection, equal to the angle of incidence.

We see that the four rays *converge* to form a real image that is inverted and reduced in size, shown here as a brown arrow. The image is at the **image distance**, q.

A **real image** is

an image formed by converging rays.

An **inverted image** is

an image that is upside down relative to the object being imaged.

A **reduced image** is

an image that is smaller in size than the object being imaged.

The **image magnification** is

the size of the image relative to the size of the object.

Note: All images formed with small concave spherical mirrors are real and inverted, unless the object is placed closer to the mirror than the focal point (p < f). Convex mirrors, by contrast, always form virtual, upright, reduced images.

Note: Real images formed with small concave spherical mirrors can produce reduced, enlarged or one-to-one (1:1) magnifications.

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How then do we determine the magnification of the image formed by ray tracing?

We could do so by actually measuring the image size from the ray tracing drawing, but a much simpler, algebraic way is available.

We first use the geometrically derived **spherical mirror equation** to get the image distance q, which states that

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q} \qquad (10.3)$$

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Again, f is the focal length of the spherical mirror, p is the object distance, and q is the image distance (all these distances are measured with respect to the vertex of the spherical mirror).

For the example depicted above,

$$\frac{1}{20} = \frac{1}{50} + \frac{1}{q}$$

We see that

$$\frac{3}{100} = \frac{1}{q}, \quad \text{or} \quad q = 33.3 \text{ cm}$$

To determine magnification algebraically, we use the **magnification equation**:

$$M = \frac{|q|}{p} \text{ x } 100 \tag{10.4}$$

Where M is the magnification of the image, expressed as a percentage, and |q| is the absolute value of the image distance q.

For the example depicted above,

$$M = \frac{|33.3|}{50} \text{ x } 100 = 66.67\ \%$$

Note: In some texts, there is concern about the signs of the distances f, p and q, but here, all distances measured to the left of the vertex of the concave spherical mirror are considered positive.

Note: The spherical mirror and magnification equations are determined from geometric considerations, and their derivations can be found online.

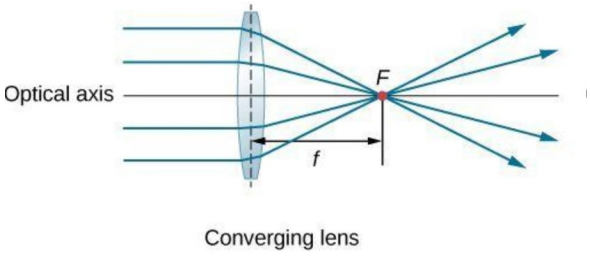
Note: When the object is placed closer to the vertex of the mirror than the focal length (when p < f), the reflected rays *diverge*. As such, they cannot be observed after reflection off a screen, like a real image formed by converging rays. Instead we observe these diverging rays by collecting them with our eyes (by looking back at the mirror).

Because the rays in this case are diverging, the image is said to be virtual. It is upright and enlarged. The image formed is like that of a magnifying lens.

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Now let's look at image formation using a lens. We will use a thin **double convex lens**:

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Converging lens

The lens is double convex because both sides curve outward (are convex).

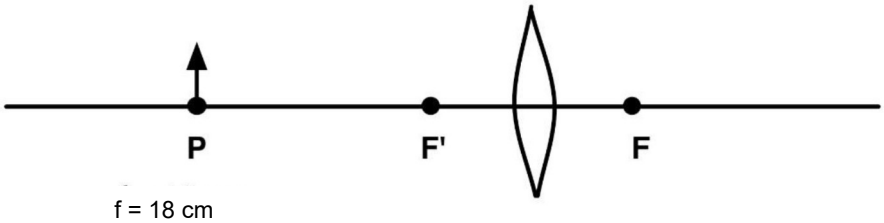
Rays coming from the left side of the lens, and parallel to the axis of symmetry (optical axis) of the lens, the rays called paraxial rays, converge to a focal point F on the exit side (or right side) of the lens.

This lens obeys the exact same image formation equation as the spherical mirror equation, and is called the **thin lens equation**:

$$\frac{1}{f}=\frac{1}{p}+\frac{1}{q} \qquad (10.3)$$

where f is the focal length of the lens (measured positive to the right of the lens in this picture), p is the object distance (measured positive to the left of the lens in this picture), and q is the image distance (measured positive to the right of the lens in this picture).

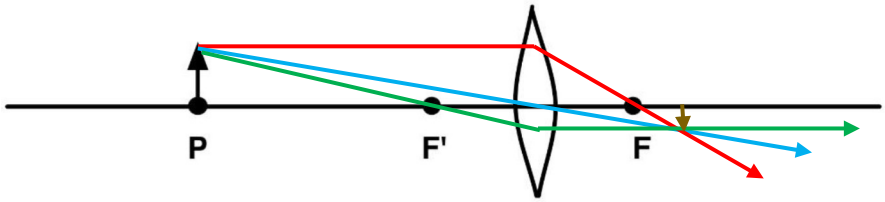
Consider the double convex thin lens shown below. The object is placed at point P, a distance p away from the lens' center, with F the focal point of the lens at distance f from the lens' center. A second (front) focal point F' is shown on the left or entrance side of the lens.



f = 18 cm

p = 54 cm

We can ray trace an image using up to three rays:



Ray 1 is the red ray, and comes in paraxially, is double refracted on each side of the lens, and goes through the focal point F. Ray 2 is the light blue ray, and goes through the center of the lens where the two sides of the lens are essentially parallel, and so goes through the lens essentially undeflected. Ray 3 is the green ray, and goes through the front focal point F' on the left or input side of the lens, and so comes out of the lens paraxially, because the reverse path of the ray would be focused at F'.

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We see that the three rays *converge* to form a real image that is inverted and reduced in size, shown here as a brown arrow.

The image formed is real, inverted and reduced in size. It can be viewed in reflection off of a screen.

Using equation (10.3), we see that

$$\frac{1}{18}=\frac{1}{54}+\frac{1}{q} \qquad (10.3)$$

and

$$\frac{2}{54}=\frac{1}{q}, \quad \text{or} \quad q=27 \text{ cm}$$

The thin double convex lens also obeys the **magnification equation**:

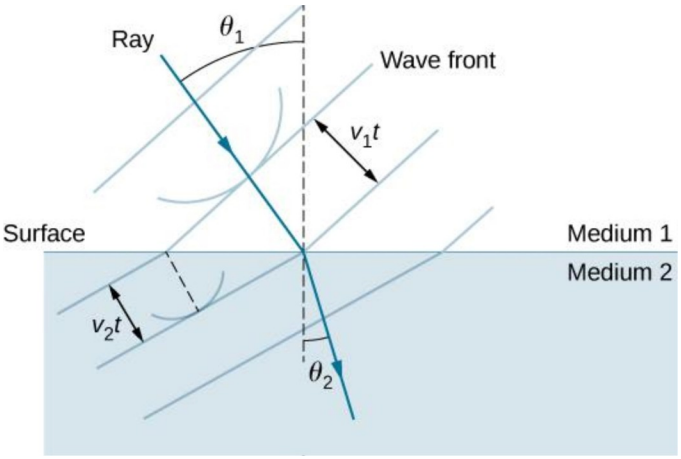
$$M=\frac{|q|}{p} \text{ x } 100 \qquad (10.4)$$

So the magnification is

$$M=\frac{|27|}{54} \text{ x } 100=50\ \%$$

Physical Optics

In order to better understand how lenses work, it is necessary to investigate refraction scientifically.



First, we observe that the ray and wave front models of light are *pictorially connected*.

Rays are straight lines that are always perpendicular to an advancing wave front (connected crest of a wave). For a plane wave front (a straight line in two dimensions), the associated rays are all parallel to each other.

Now we know, experimentally, that refraction occurs because light changes speed as it crosses a boundary between two optical (transparent or translucent) materials of different optical densities, shown here as medium 1 and medium 2.

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In time t, the wave front advances v_1t distance, where v_1 is the speed of light in medium 1.

In the same time t, the part of the wave front already in medium 2 advances a shorter distance v_2t, where v_2 is the speed of light in medium 2.

The curved lines in the image above are a depiction of **Huygens' Principle**, which states that

> each point on a wave front can be considered a point source of circular wavelets.

Now in trying to understand how a wave front advances, we use both **Huygens' Principle** and the **Principle of Superposition** (covered in Module 7) to justify that the plane wave front ***stays*** a plane wave front upon entering medium 2, but the plane wave front has to bend, because light changes speed when entering the second medium.

For this example, medium 2 is denser than medium 1, and v_2 is slower than v_1.

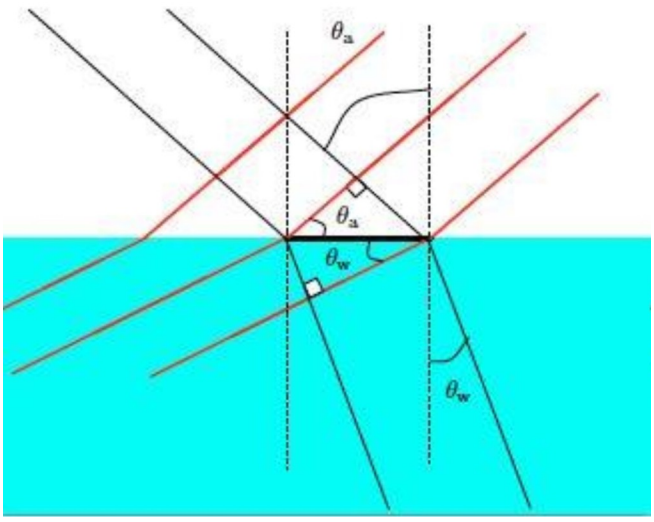
We see that the light blue ray of interest bends in towards the dotted perpendicular line (the **normal**) when entering medium 2, because light moves more slowly in medium 2 than medium 1.

In general, we know from experiment that rays are bent

> in towards the normal when going from one optical medium to another of a higher optical density, and are bent out from the normal when going from one optical medium to another of a lesser optical density.

We are now ready to mathematically state how rays of a plane wave are bent, in going from one optical medium to another of a different optical density.

In the image below, the red plane wave front is refracted at a boundary between air (a) and water (w).



If we look at just the left-most black ray, we see that it is bent in towards the normal (dotted line), as expected (water is optically denser than air).

The left side of the heavy black line indicates where the wave front entered the water at time t = 0 sec. The right side of the heavy black line indicates where the wave front entered the water at time t = t sec.

The angle the plane wave front segment in the water makes with the heavy black line is θ_w, and the angle the plane wave front segment in the air makes with the heavy black line is θ_a.

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The side opposite θ_w has length v_2t (from the previous illustration), and the side opposite θ_a has length v_1t (from the previous illustration).

If the length of the heavy black line is d, we see that

$$\sin \theta_w = v_2t / d, \text{ and } \sin \theta_a = v_1t / d$$

Combining the two expressions, we see that

$$\sin \theta_w \ / \sin \theta_a = v_2 / v_1 \text{ , or}$$

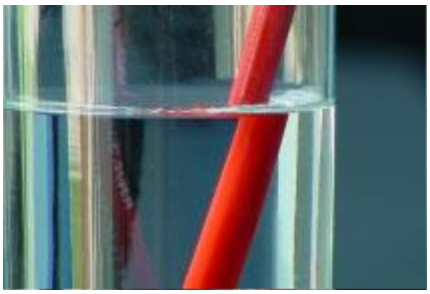
$$\sin \theta_w \ / v_2 = \sin \theta_a / v_1$$

Multiplying both sides of this equation by the speed of light in the vacuum, c, we get

$$(c / v_2) \times \sin \theta_w = (c / v_1) \times \sin \theta_a$$

Identifying the index of refraction n of a material as c divided by the speed of light in that material, and identifying θ_w as θ_2, and θ_a as θ_1, we arrive at **Snell's Law of Refraction**:

$$n_1 \times \sin \theta_1 = n_2 \times \sin \theta_2 \qquad (10.5)$$



Refraction is abundant in nature. The corneas and lenses in healthy eyes refract light so as to create real, inverted and reduced images in focus on the retinas (light sensitive screens) in the back of our eyes.

The image to the left shows what appears to be a broken pencil, but what we see is an optical illusion. The bottom part of the pencil is undergoing two refractive events: water to glass, and glass to air. The top part of the pencil is also undergoing two refractive events, but they are different: air to glass, and glass to air. Keep in mind that the curvature of the glass is also contributing to this optical illusion.

Indices of refraction for common optical materials can be found here.

*****

We can also use **Huygens' Principle** and the **Principle of Superposition** to explain the classical light phenomenon of diffraction.

Recall that diffraction

is the bending of light around corners.

In diffraction, there is no change of medium. Particles of light cannot diffract, as they go only in straight lines.

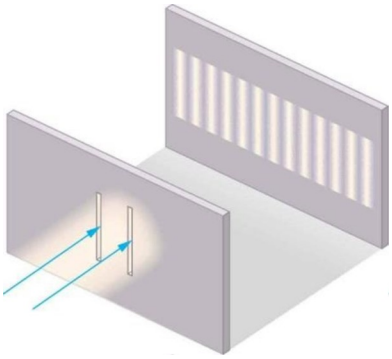
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Diffraction occurs for all light sources, whether visible or invisible, whether a single frequency or a mixture of frequencies, but coherent light, specifically (visible) laser light, is ideal for demonstrating diffraction.

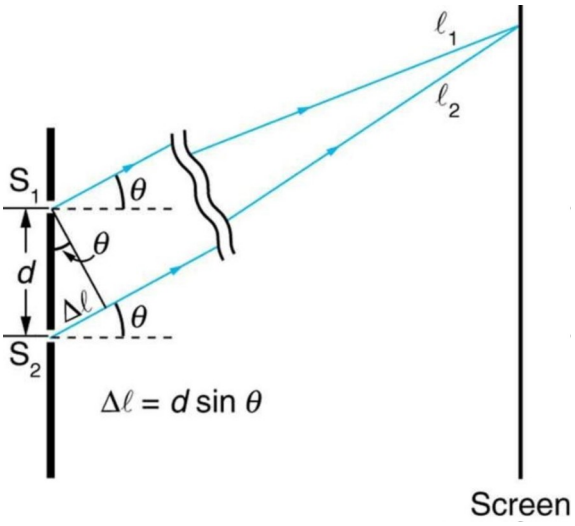
Diffraction as a wave phenomenon was solidly confirmed in 1801 by an Englishman named Thomas Young. In his classic experiment, he sent a plane wave front through an aperture plate having two narrow slits in it.

The result was that he saw an interference pattern on a screen parallel to the aperture plate but a significant distance away, a clear and indisputable indication that light is a wave.



In the image to the left, we see a simplified picture of what Young did in his classic experiment.

The bright spots, called maxima, are locations where the "two point sources of circular wavelets" coming from the two aperture slits interfere constructively. The darker spots in between, called minima, are where the "two point sources of circular wavelets" coming from the aperture slits interfere destructively.



Mathematically, the two slits, separated by a small distance d (on the order of the wavelength of the coherent light being diffracted), send nearly parallel rays to a point off the **axis of symmetry** of the diffraction apparatus (a horizontal line between the midpoint of the line segment d and the "far off" screen).

The two "point source wavelets" from the coherent laser light source are in phase (both having crests at the same time) when they start out from their respective slits S_1 and S_2, but the ray from slit S_2 has farther to travel.

If that distance, here indicated as $\Delta\ell$, is an integer multiple of the light's wavelength, a bright spot or maximum will occur, i.e.,

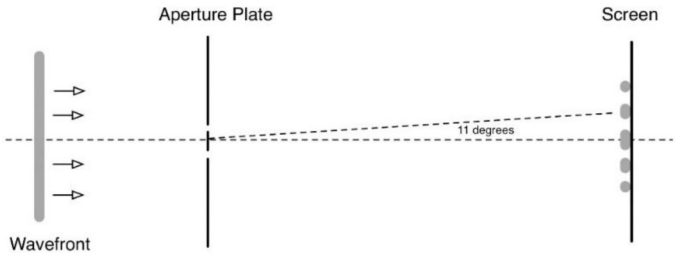
$$d \sin\theta = n\lambda \qquad (10.6)$$

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where n is a positive or negative whole number, including zero.

The bright spot (maximum) on the axis of symmetry corresponds to n = 0, whereas the maxima above the axis of symmetry correspond to positive integer values of n. Maxima below the axis of symmetry correspond to negative integer values of n.

Consider the two slit apparatus pictured below. A plane wave front of coherent light is striking the aperture plate, causing a diffraction pattern on the screen far away.



The separation distance d between the two slits is given as 2.21×10^{-6} m, and the first maximum off the axis of symmetry is measured to be an angle of 11°. What is the wavelength of the coherent light?

From equation (10.6), we see that

$d \sin \theta = n \lambda = (2.21 \times 10^{-6}) (\sin 11) = 1 \times \lambda = 0.422 \times 10^{-6} = 422$ nm, a violet laser.

We will do a lab activity involving diffraction of coherent laser light.

*****

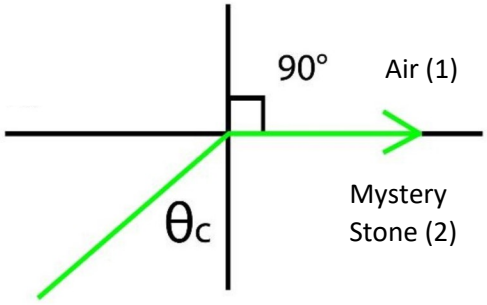
First Extended Study: If a light ray travels from one optical material to another of a lesser optical density, the ray is “bent out from the normal” provided that the angle predicted by Snell’s Law is less than or equal to 90° (equivalently has a sine less than or equal to 1).

If Snell’s Law predicts that the refracted ray has an angle of refraction whose sine is greater than 1, the ray cannot escape the denser medium, and is reflected instead.

There is always a critical angle for the denser medium, such that any ray having an angle of incidence greater than that critical angle will always be reflected. If a beam of laser light in the denser medium has such an incident angle, it will be **totally internally reflected**.

Consider this example.

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A young woman receives an engagement ring with a large stone in it.

She is suspicious that her fiancé has given her not a diamond, but a cheap substitute, namely a cubic zirconia stone.

Using her iPhone Whatever, she determines that the index of refraction for a diamond is about 2.42, whereas the index of refraction for a cubic zirconia stone is around 2.18.

Curious, she devises an experiment to find out what the critical angle (the angle at which total internal reflection occurs) of her mystery stone is, relative to air. She finds that that angle (θ_c) is 24 degrees.

What is the index of refraction of her engagement ring's mystery stone? Is her fiancé the real deal, or is he a cheap bum?

To solve this problem, we must use equation (10.5):

$$n_1 \times \sin \theta_1 = n_2 \times \sin \theta_2 \qquad (10.5)$$

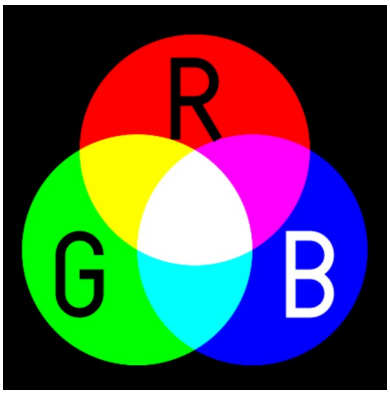
Since the speed of light in air (medium 1) is very close to the speed of light in the vacuum of space, we can set n_1 to 1. Since the refracted ray is at 90°, we set $\sin \theta_1$ to 1 as well. Since θ_2 is θ_c and is equal to 24°, we see that

$$(1)(1) = n_2 \sin 24 = n_2 (0.4067) \text{ , or}$$

$$n_2 = 2.458, \text{ and the mystery stone is a diamond.}$$

*****

Second Extended Study: Earlier in this module, we stated that the colors we see in visible light are biological artifacts, not objective qualities of the light itself. Let's explore this further.



When lighting is dim, human vision is exclusively in black and white. Rods in the retina of our eyes are responsible for this kind of vision.

But when lighting is of normal daytime intensity, humans experience color vision. The colors that we see add an enormous amount of vital information about our surroundings.

In the retina of our eyes are **three kinds of color cells** called cones. They are named according to which colors they are most responsive to, namely red (R), green (G) and blue (B).

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It is important to realize that the red cones see not just a specific frequency of light we call red, but a range of frequencies centered around a dominant frequency we call red. It follows that a red cone and a green cone might respond, albeit in differing ways, to a single shared frequency of visible light.

Red cones (L cones, for long wavelength) respond best to wavelengths around 565 nm, but their sensitivity reaches farthest into the red end of the visible spectrum.

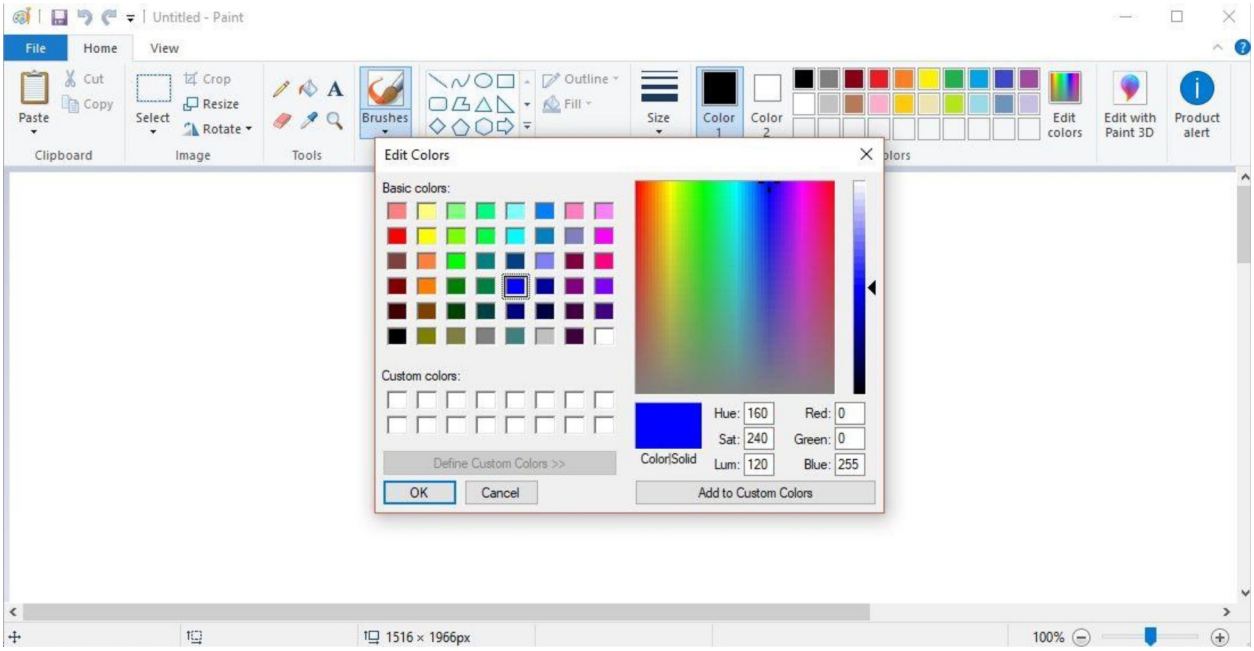
Blue cones (S cones, for short wavelength) respond best to the shortest wavelengths in the visible spectrum, around 430 nm.

Green cones (M cones, for medium wavelength) respond best to wavelengths in the middle of the visible spectrum, around 535 nm.

All the colors we see in the rainbow of colors are ultimately a combination of intensities of the **primary colors**, namely pure red, pure green and pure blue.

You can easily verify this by opening “Paint” in most any current version of Windows, and selecting “Edit Colors.”

If you choose the blue shown in the image below (fourth color down in the fourth row), you will see a full-intensity blue (255 on a scale of 0 to 255), with no red (0 chosen) and no green (0 chosen).



If you instead choose the second color down in the first column, you will see a full-intensity red (255), with no blue (0 chosen) and no green (0 chosen).

If you choose the third color down in the third column, you will see a full-intensity green (255), with no blue (0 chosen) and no red (0 chosen).

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Note that for the additive color theory of light, the “color” white is full-intensity red (255) plus full-intensity green (255) plus full-intensity blue (255).

Black by contrast is no red (0), no green (0) and no blue (0).

Of interest to us here are the **complementary colors** shown between the primary (RGB) colors on the “color wheel” shown above.

The color *yellow* is full-intensity red (255) plus full-intensity green (255), with no blue (0). Yellow is considered the **complement** of blue. We say that yellow is white light minus blue.

Similarly, the color *cyan* is full-intensity green (255) plus full-intensity blue (255), with no red (0). Cyan is considered the **complement** of red. We say that cyan is white light minus red. Think of cyan as turquoise.

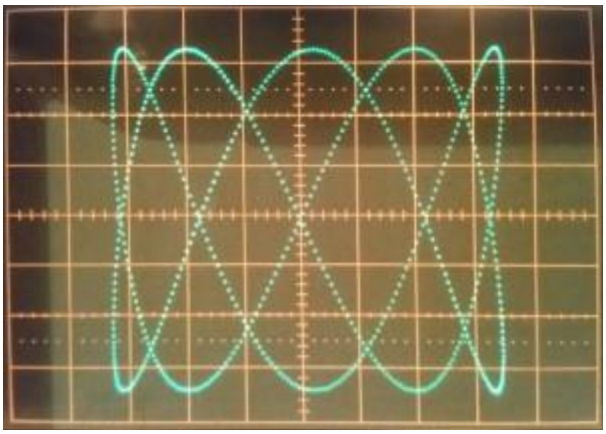
Finally, the color *magenta* is full-intensity red (255) plus full-intensity blue (255), with no green (0). Magenta is considered the **complement** of green. We say that magenta is white light minus green. Think of magenta as a purplish red, a somewhat royal color.

Note: Colors have deep psychological significance for most people. Click here or here for further information on this fascinating subject.

Note: If you are wondering what the color brown is, it is a half-intensity red (128), with a little green (64) and no blue (0).

HOMEWORK

1. What is the Frequency Kenneth?



The wavelengths of visible light range from approximately 700 nm (dark red) to 400 nm (dark violet). With the speed of light given as 3×10^8 m/sec, use equation (10.1) to determine frequencies of light corresponding to these wavelengths.

Answers: Dark red has a frequency of 4.286×10^{14} Hz, and dark violet has a frequency of 7.5×10^{14} Hz.

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2. Who Is This Hertz Fellow?



Heinrich Hertz was a German physicist best known for being the first person to create and detect an electromagnetic wave.

James Clerk Maxwell had postulated the existence of such a wave, but no one knew for sure if it existed, until Hertz successfully showed in 1886 that it did. The frequency his first radio transmitter produced was an astounding 50 MHz (million Hertz).

Ever the keen observer, Hertz was the first to notice that sparks jumped more readily across a gap exposed to UV light (a year later, Wilhelm Hallwachs showed that UV light makes charged metal plates lose their charge). This observation became known as the photoelectric effect, which was ultimately explained by Albert Einstein in 1905.

Hertz died in 1894, at the ripe young age of 36, less than a decade before the Nobel Prizes were first given. In 1960, the unit “cycles per second (cps)” was rechristened Hertz in his honor.

3. How Fast Does Light Go Again?

The speed of light in the vacuum of outer space is an astoundingly large number, roughly 3×10^8 m/sec.

Light moves so quickly that it takes less than 1.5 seconds to come from the Moon to Earth.



When the Apollo astronauts were on the Moon, Houston (Mission Control) would call out to the astronauts on the Moon, only for there to be an awkward delay of about three seconds before anyone heard a response.

If you could travel at the vacuum speed of light, you could circle the Earth 7.5 times per second.

If you could travel at the vacuum speed of light, you could travel to the Sun in 500 seconds, or 8.33 minutes.

It’s important to realize that when light travels through an optical medium, it slows down, sometimes considerably.

The **index of refraction** of an optical medium x (called n_x) is the speed of light in the vacuum (c) divided by the speed of light in the medium (v_x).

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We write

$$n_x = c / v_x \qquad (10.7)$$

For example, the index of refraction for water is 1.33, meaning that light travels at a speed of 3/4ths of c in water.

*****

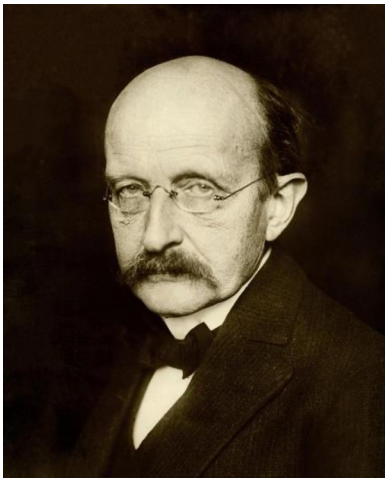


Just because material particles are forbidden from traveling at (or in excess of) c under any circumstances doesn't mean that material particles can't travel faster than light when moving through an optical medium such as water.

When electrons move at speeds less than c but greater than the speed of light in the optical medium being used, those electrons will give off a special kind of light called Cherenkov radiation (as in the image to the left). This radiation is akin to the sound shock wave (sonic boom) that occurs when planes travel faster than the speed of sound in air.

4. Mr. Planck and His Hypothesis

In December of 1900, the end of the first year of a new century, the German Physicist Max Planck proposed a solution to a vexing problem in physics dubbed "the ultraviolet catastrophe."



Classical wave-based electromagnetic theory predicted that the energy emitted by a heated, light emitting "blackbody" would veer to infinity for high (as in ultraviolet) frequencies of light, an obviously non-sensical conclusion.

Planck's solution, called **Planck's hypothesis**, was to propose that electromagnetic energy was emitted, in blackbodies, by charged particles (electrons) in a "chunky" or quantized fashion. He proposed that blackbodies emitted light according to the formula

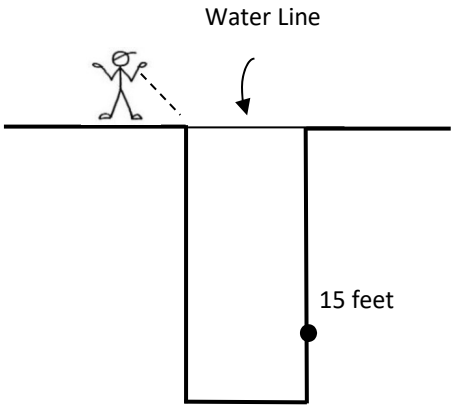
$$E = h \times f \qquad (10.8)$$

Where E is the energy emitted in a single quantum (chunk) of light, f is the frequency of the light emitted, and h was a constant that would come to be known as Planck's Constant, namely 6.626×10^{-34} Joule-sec. Planck himself attributed this quantization to the emitting electrons; it was Albert Einstein, in 1905, who proposed that light itself comes in particles, later called photons.

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Planck's hypothesis would become the basis for a whole new canon of modern physics called quantum mechanics.

5. Swimming With The Fishes (Advanced Students Only)



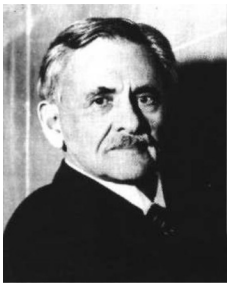
A swimming coach stands 5 feet from the edge of a swimming pool.

His eyes are 5 feet off the floor.

Looking down and just over the near side edge of the pool, the lowest depth marker he can see on the opposite side of the pool is the 15 foot marker. How wide is the pool? (Use n = 1.33 for water.)

Answer: 9.412 feet

6. The Michelson-Morley Experiment



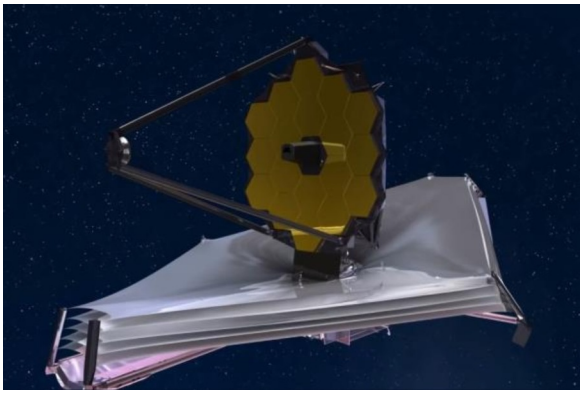
The constancy of the speed of light in air (and ultimately in the vacuum) was established experimentally in 1887 by two Americans, Albert Michelson and Edward Morley, who utterly failed to find the mysterious but presumed substance, called the aether, that supposedly transported light from one place to another.

The experiment was so thorough, and was performed, as a check, at different times of the year, that the scientific community was eventually forced to conclude, especially after Einstein's special relativity (1905), that the aether didn't exist, and that light traveled through apparently empty space with no need for a medium. Put another way, they were forced to conclude that light was a non-mechanical wave.

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ASTRONOMY SUPPLEMENT

1. Light and Time



Almost everything we know about the Universe comes from studying light that is coming to us from glowing objects, or collections of glowing objects.

For close-in objects like planets, we rely on reflected light, but stars nearby glow with their own light in all the typical frequency ranges.

Farther away, we see only collections of stars, or galaxies, and even farther away, we see collections of galaxies.

The light may be visible, or invisible. Regarding invisible light, currently we have satellites that collect light in the infrared, ultraviolet and x-ray ranges of light frequencies.

As we look at light from collections of glowing objects farther and farther away, we ineluctably see further and further backwards in time. This is because of light's finite speed of 186,000 miles per second. The Hubble Space Telescope has imaged galaxies whose light left them more than 13 billion years ago.

Since 2022, astronomers have been observing infrared light, with extreme precision, using the James Webb Telescope, which has seen galaxies as they were only a few hundred million years after the beginning of our Universe (about 13.8 billion years ago).

Click on the image above to see what this telescope looks like, and what it has discovered so far.

2. Measuring the Speed of Light



The speed of light has been a fascination for scientists for centuries.

The first person to attempt to measure the speed of light was a Danish astronomer named Ole Romer, who in 1676 reported timing the passage of the moons of Jupiter behind the planet at different times of the year. His data led Christiaan Huygens to an estimate of 220,000 kilometers per second, a far cry from **light speed's current known value** of

299,792.458 kilometers per second

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This speed is important in astronomy precisely because *it is a constant*. In fact, the value of the speed of light (in the vacuum of outer space) is a *law of physics*.

We denote the speed of light (in the vacuum of outer space) by the letter c, and in calculations approximate its value as

c = 300,000 kilometers per second = 186,000 miles per second

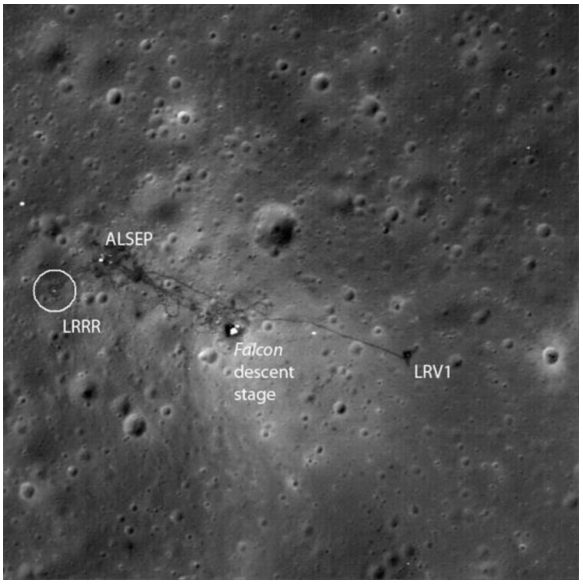
The next person after Romer to attempt a calculation of light's speed was the English astronomer James Bradley, who in 1728 used a technique called stellar aberration to determine that c was approximately 301,000 kilometers per second. This is an astonishingly good result, considering he calculated it almost 300 years ago.

In the 19th century a pair of Frenchmen measured the speed of light in the laboratory. In 1849, a Frenchman named Fizeau used a rotating gear and mirror system to determine light's speed (in air) to be 313,000 kilometers per second.

Later, in 1862, another Frenchman, Leon Foucault, used a rotating and fixed mirror system to estimate light's speed (in air) to be 298,000 kilometers per second, within 0.6% of the modern value.

Finally, in the 20th century, the American physicist Albert Michelson used an extended vacuum tube variation of Foucault's method to determine that the speed of light (in a near vacuum) was 299,774 kilometers per second.

3. Lunar Laser Ranging



To this day, coherent light from powerful Earth based laser beams are sent to the Moon and reflected back to Earth to measure how far away the Moon is at the time of measurement.

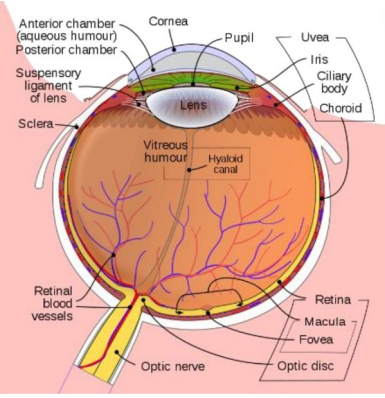
There are five classic reflector arrays on the Moon, three left by Apollo astronauts (from the Apollo 11, 14 and 15 missions), and two left by Soviet robotic rovers (Lunokhod 1 and 2). Smaller reflectors have since arrived on newer landers.

We know the distance to the Moon to millimeter precision. Currently, the Moon is migrating away from the Earth about 3.8 centimeters per year.

4. Collecting and Imaging Light – Telescopes

There are many devices that collect light, and in doing so, distort the light in such a way as to form an image. The image may be merely viewable, but also may be permanently

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recordable. We learned about image formation with mirrors and lenses earlier in this module.

Our eyes collect light stereoscopically: two different upside down images (one per eye), in a combination of color and black and white lighting, are "combined" in each viewer's brain to form a three-dimensional vision of reality. The color portion of the image is created electrochemically in the eye's retina, using the three **color cones** (RGB) mentioned in the **Second Extended Study**.

A reasonable facsimile of a human eye is the camera.



Using the lens imaging ideas learned earlier in this module, the camera captures single (or still) images, or multiple sequential images (a movie or video), and saves them either in (now the old fashioned) film format, or the (current and modern) digital format. Digital images are captured by electronic image sensors: CMOS sensors in phones, and charge coupled devices (CCD) in many scientific cameras.

Most every phone manufactured today has this digital imaging technology, and many smart phones allow elementary editing of digital images.

In astronomy, the primary light collecting and imaging devices are binoculars and telescopes.



Binoculars magnify a distant object in such a way that it seems much closer. Some binoculars inadvertently (or on purpose) effectively separate a person's two eyes horizontally so as to give that person an enhanced sense of depth.

Modern binoculars fold the optical paths of the two sides of the device using prisms. This ensures that the image seen is upright and appropriately bright. For binoculars with large enough primary (or entrance) lenses, the viewer can easily see details on the Moon's surface.

Telescopes, whether for professionals or amateurs, come in **two basic forms**: refractors and reflectors.

The primary light collecting and shaping device in a **refractor telescope** is its primary lens. Because of the weight (and subsequent sagging) of a primary lens, such lenses

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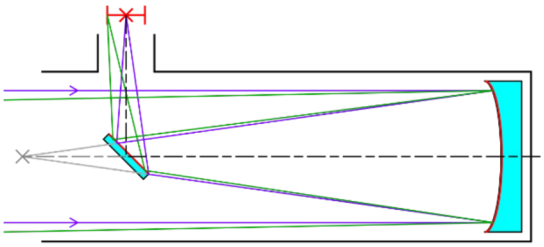
can only be so big. The largest professional refractor telescope is the 40 inch diameter

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refracting lens at the Yerkes Observatory. The Yerkes Observatory closed in 2018, but a nonprofit foundation reopened it to the public in 2022.

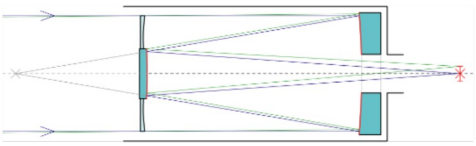
It follows that all large modern research telescopes are reflector telescopes (small refractors remain popular with amateurs).

The quality of a **reflector telescope** is measured by its primary mirror. The larger the primary mirror, the better the mirror's light collecting ability, and the better the mirror's ability to separate glowing objects that are angularly closely spaced.



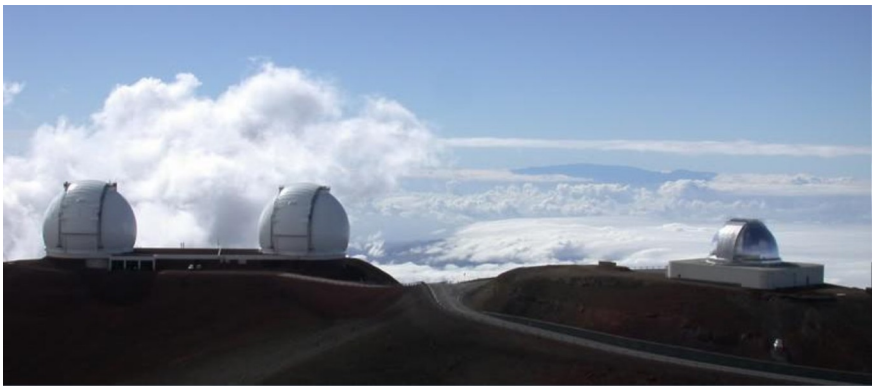
Sir Isaac Newton built the first practical reflector telescope, called the Newtonian Reflector. A small flat mirror is placed in the middle of the primary concave mirror's optical path, reflecting the intercepted light to a focal point just inside the focal point of the telescope's eyepiece, mounted on the side of the telescope (red x area).

An amateur version of this telescope is the Dobsonian, which has a large primary mirror mounted on an inexpensive and (sometimes) crude "alt-azimuth" base. Primary mirror diameters for Dobsonians commonly range from about 6 to 20 inches or more.



A better but more expensive (amateur) reflector telescope is the **compound telescope** (or catadioptric telescope), which involves a large primary mirror whose optical path is "folded" by a second internal reflecting mirror that effectively shortens the tube length of the telescope, and allows it to be mounted on a more sophisticated, clock drivable "equatorial" base. The entrance glass (corrector plate) of such scopes is shaped to correct the spherical aberration of the primary mirror.

There are two popular compound telescopes, and variants of them: the Schmidt-Cassegrain Telescope, and the Maksutov-Cassegrain Telescope. Primary mirrors in compound telescopes typically run from 5 to 14 inches in diameter.



Ground based professional reflector telescopes, no matter the form of the optical path, have much larger primary mirrors, like the 8.2 meter (323 inch) Subaru Telescope, situated on the top of Mauna Kea.

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Mauna Kea is one of the five volcanoes that make up Hawaii's Big Island. Its summit, at 13,803 feet, sits above the inversion layer, about 40% of the atmosphere, and most of its water vapor. Pictured above are the 2 ten meter diameter Keck telescopes, also on Mauna Kea, which are each composed of 36 hexagonal mirrors, kept precisely aligned by active optics. A separate adaptive optics system corrects for air turbulence to produce clear images.



The best way to completely avoid imaging errors associated with air turbulence (and man-made "light pollution") is to put astronomical observatories in outer space.

Some **space-based observatories** are placed at Lagrange points, some orbit the Sun like another planet, most orbit the Earth, and at least one, the Parker Solar Probe, directly orbits the Sun close up.

These space-based telescopes vary as to which frequencies they dominantly receive and use for imaging.

Notable space-based telescopes are the

Swift Explorer, which monitors gamma ray, x-ray and visible light sources,

Fermi Space Telescope, which monitors gamma ray light sources,

Chandra Observatory, which monitors x-ray light sources,

Hubble Space Telescope, which monitors ultraviolet, visible and infrared light sources, and the

James Webb Space Telescope, which monitors infrared light sources (succeeding the Spitzer Space Telescope, retired in 2020).



The most notable of these space-based telescopes is the **Hubble Space Telescope (HST)** pictured above. Originally launched in 1990, and after five astronaut servicing missions, it is still operating today, producing an amazing gallery of images in the visible, ultraviolet and infrared frequency ranges. A particularly notable achievement of the HST is its deep field image (shown on the left), which showed how the Universe appeared more than 12 billion years ago.

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To finish up, a brief mention of Earth-based radio telescopes is in order: while they produce imprecise images (due to the long wavelengths of radio signals from space), such telescopes can see through interstellar dust, a notable advantage.



Important radio telescopes include the Green Bank Telescope in West Virginia, the Arecibo Observatory in Puerto Rico (which collapsed in 2020), and the Very Long Baseline Array, ten dishes spread from Hawaii to the U.S. Virgin Islands and operated from Socorro, New Mexico.

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